

Problems

- ① Experience has shown that 20% of manufactured products are of top quality. In today's production of 400 article only 50 are of top quality. Show that either the production of the day chosen was not a representative sample or hypothesis of 20% ^{was} wrong. Based on the particular day production, find also 95% Confidence limit for the percentage of top quality product.

Solⁿ ● $H_0: P = \frac{20}{100} = \frac{1}{5}$

$$Q = 1 - \frac{1}{5} = \frac{4}{5}$$

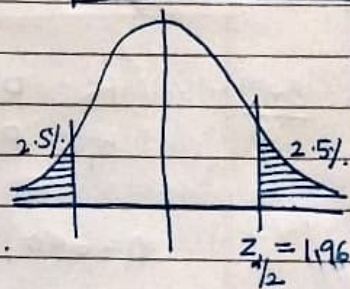
ie $H_0: P = \frac{1}{5}$

$H_1: P \neq \frac{1}{5}$ (two tailed test)

$p = \frac{50}{400}$

Sample size $n = 400$, $p = \frac{1}{5}$, $\alpha = 5\%$
using z-test

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$$



$$Z = \frac{\frac{1}{8} - \frac{1}{5}}{\sqrt{\frac{\frac{1}{5} \times \frac{4}{5}}{400}}}$$

ie $Z = \frac{-15}{4} = -3.75$

here $Z_{\alpha/2} = 1.96$, $Z_{cal} = -3.75$

as

$$|Z_{cal}| > |Z_{\alpha/2}|$$

ie Null Hypothesis is rejected. ie alternate hypo. H_1 is accepted. ie difference b/w population proportion and sample proportion are significant.

95% confidence limit for top quality product

$$\text{from } |Z| \leq 1.96$$

$$\text{ie } \left| \frac{p - P}{\sqrt{\frac{Pq}{n}}} \right| \leq 1.96$$

$$p - 1.96 \sqrt{\frac{Pq}{n}} \leq P \leq p + 1.96 \sqrt{\frac{Pq}{n}}$$

$$\text{ie } \frac{1}{8} - 1.96 \sqrt{\frac{1 \times 7}{8 \times 400}} \leq P \leq \frac{1}{8} + 1.96 \sqrt{\frac{1 \times 7}{8 \times 400}}$$

$$0.093 \leq P \leq 0.157$$

- ② The fatality rate of typhoid patient is belived to be 17.26 percent. In a certain year 640 patient suffering from typhoid were treated in a metropolitan hospital and only 63 patient died. can you consider hospital efficient at 1% level of significance?

Soln

$$H_0: P = 0.1726$$

$$H_1: P < 0.1726 \quad (\text{one tailed test, left tailed})$$

$$n = 640, \quad Q = 1 - 0.1726 = 0.8274$$

$$p = \frac{63}{640} = 0.0984$$

Using Z-Test

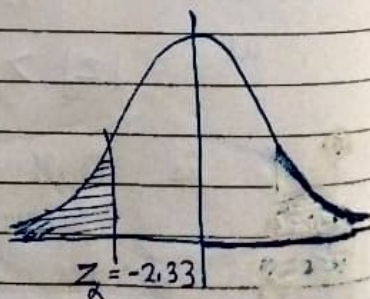
$$Z = \frac{p - P}{\sqrt{\frac{Pq}{n}}} = \frac{0.0984 - 0.1726}{\sqrt{\frac{0.1726 \times 0.8274}{640}}}$$

$$Z_{\text{cal}} = 4.96$$

Now let level of significance is 1%.

From Table

$$Z_{\alpha} = Z_{0.01} = -2.33$$



here $|Z_{cal}| > |Z_{\alpha}|$

ie H_0 is rejected and H_1 is accepted.
ie difference b/w sample proportion and
population proportion is significant
& Hospital is efficient.

Q:- A manufacturer claimed that at least 95% of the equipment which he supplied to a factory conformed to specification. An examination of a sample of 200 pieces of equipment revealed that 18 were faulty. Test this claim at 5% level of significance.

Soln:- $P = \frac{95}{100} = .95$

H_0 &

- ① Null Hypothesis $H_0: P = .95$ ie proportion of pieces conforming to proportion is 95%
- ② Alternate Hypothesis $H_1: P \neq .95$ (left tailed)

$n = 200, p = \frac{182}{200} = 0.91, \alpha = 5\%$

$Q = 1 - P = 1 - .95 = .05$

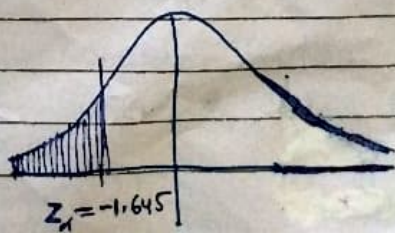
③ Test Statistic

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{.91 - .95}{\sqrt{\frac{.95 \cdot .05}{200}}} = -2.59$$

In Sample
200-18 = 182
items are
conformed to
specification

$|Z_{cal}| = 2.59$

$|Z_{cal}| > |Z_{\alpha}|$ ie H_0 is
rejected.



- ie H_0 is accepted.
 ie manufacturer's claim is rejected.

Q:- A salesman in departmental store claims that at most 60% of the shoppers entering the store leave without making a purchase. A random sample of 50 shoppers showed that 35 of them left without making a purchase. Are these sample results consistent with the claims of salesman. Use l.o.s of .05.

Soln:- ① Null Hypothesis $H_0: P = \frac{60}{100}$

$$H_0: P = \frac{3}{5} = .6$$

② Alternate Hypothesis

$$H_1: P \geq .6 \quad (\text{right tailed test})$$

$$n = 50, Q = 1 - P = .4, p = \frac{35}{50} = .7$$

③ Test Statistic

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{.7 - .6}{\sqrt{\frac{.6 \times .4}{50}}}$$

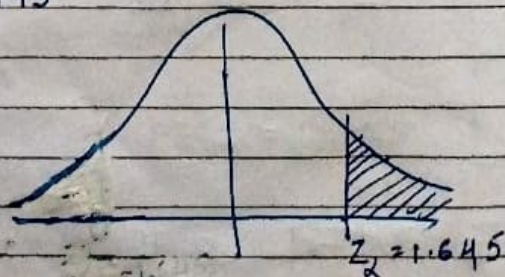
$$Z = 1.443$$

given

$$\alpha = .05$$

ie from table

$$Z_{\alpha} = 1.645$$



$$|z_{cal}| < |z_{\alpha}|$$

ie Null Hypothesis is accepted.

Q: If 40 people were attacked by a disease and only 32 survived, will you reject the hypothesis that the survival rate if attacked by the disease is 85%, in favour of the hypothesis that it is more than at 5% level.

Soln:- proportion of survival rate = 85%.

Let $H_0: P = \frac{85}{100} = .85$

$H_1: P > .85$ (right tailed)

$Q = 1 - P = 1 - .85$

$= .15$

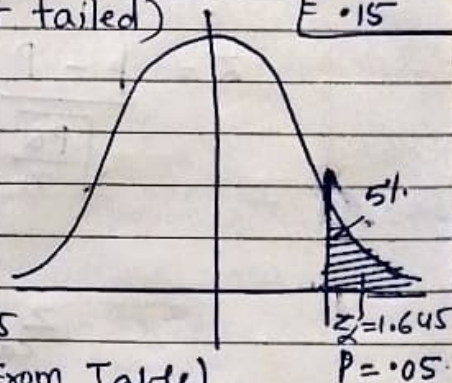
$p = \frac{32}{40} = .8$

level of significance = 5%

at 5% level of significance

critical point $z_{\alpha} = 1.645$

(From Table)



Test statistic:-

$$z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$$

$$z = .8 - .85$$

$$= -.05$$

$$\sqrt{\frac{.85 \times .15}{40}}$$

$$= .05645$$

$$= -.8857$$

$|z_{cal}| < |z_{\alpha}|$ ie Null Hypothesis is accepted.

Q:- In a sample of 1000 people in Maharashtra 540 are rice eaters and rest are wheat eaters. Can you assume that both rice and wheat eaters are equally popular in this state at 1% level of significance.

soln:- $n = 1000$, $p = \frac{540}{1000} = .54$, = proportion of rice eater in a sample of 1000.

Let both are equally popular

Let $H_0: P = 0.5$ (proportion of rice eater)

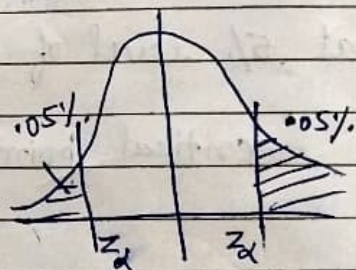
$H_1: P \neq 0.5$ (two tailed)

$$\alpha = 1\%$$

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{.54 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{1000}}}$$

$$Z_{cal} = 2.532$$

$$|Z_{cal}| = 2.58$$



$|Z_{cal}| < |Z_{\alpha}|$ i.e. H_0 is accepted

i.e. both are equally popular.

Questions for Difference of proportion :-

- Q1. Random samples of 400 men and 600 women were asked whether they would like to have a flyover near their residence. 200 men and 325 women were in favour of the proposal. Test the Hypothesis that proportion of men and women in favour are same at 5% level of significance.

Solⁿ:- $n_1 = 400$, $n_2 = 600$

$$p_1 = \frac{200}{400}, \quad p_2 = \frac{325}{600}$$

$$p_1 = \frac{1}{2} = 0.5 \quad p_2 = 0.541$$

Let Null Hypothesis is there is no significant difference b/w the proportion of men and women if favour

1) ie $H_0 : p_1 = p_2$

2) Alternate Hy. $H_1 : p_1 \neq p_2$ (two tailed)

3) $\alpha = 5\%$

4) test statistic $Z = \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$

where $P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$

$R = \dots$

$$P = \frac{400 \times 0.5 + 600 \times 0.541}{400 + 600}$$

$$P = 0.525$$

$$Q = 1 - P$$

$$Q = 0.475$$

$$Z = \frac{p_1 - p_2}{\sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$Z = \frac{0.5 - 0.541}{\sqrt{0.525 \times 0.475 \left(\frac{1}{400} + \frac{1}{600} \right)}}$$

$$Z = \frac{-0.041}{0.032} = -1.34$$

$$\text{ie } [Z_{\text{cal}} = -1.34]$$

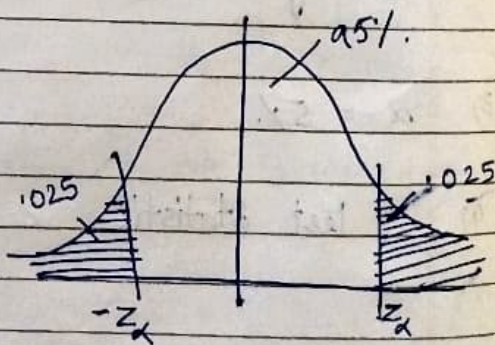
given $\alpha = 5\%$

$$\text{ie } Z_{\alpha} = 1.96$$

(from Table)

ie

$$|Z_{\text{cal}}| < |Z_{\alpha}|$$



ie Null Hypothesis is accepted at 5% level of significance. ie there is no difference

of opinion b/w men and women as far as proposal of flyover is concerned.

Q2 In a large city A, 20% of random sample of 900 school boys had a slight physical defect. In another city B, 18.5% of a random sample of 1600 school boys had the same defects. Is the difference b/w the proportion significant.

Soln

$$p_1 = \frac{20}{100} = 0.2, \quad p_2 = \frac{18.5}{100} = 0.185$$

$$n_1 = 900, \quad n_2 = 1600$$

$$P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{0.2 \times 900 + 0.185 \times 1600}{900 + 1600}$$

$$P = 0.1904$$

$$Q = 0.8096$$

Let $\alpha = 5\%$.

Null Hypo. $H_0: p_1 = p_2$

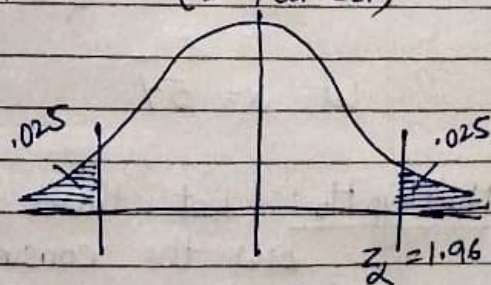
$H_1: p_1 \neq p_2$ (2 tailed)

Test statistic :-

$$Z = \frac{p_1 - p_2}{\sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$Z = \frac{0.2 - 0.185}{\sqrt{0.1904 \times 0.8096 \left(\frac{1}{900} + \frac{1}{1600} \right)}}$$

$$Z = 0.92$$



$$|Z_{cal}| < |Z_{\alpha}|$$

ie Null Hypo H_0 is accepted.

ie diff b/w proportion is not significant.

Q:- Before an increase in excise duty on tea, 800 persons out of a sample of 1000 persons were found to be tea drinkers. After an increase in duty, 800 people were tea drinkers in a sample of 1200 people. Using standard error of proportion, state whether there is a significant decrease in the consumption of tea after the increase in excise duty.

Solⁿ:

$$p_1 = \frac{800}{1000} = .8 \quad n_1 = 1000$$

$$p_2 = \frac{800}{1200} = .667 \quad n_2 = 1200$$

$$P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = 0.727$$

$$P = 0.727 \quad Q = 1 - 0.727$$

$$Q = 0.273$$

Let $\alpha = 5\%$.

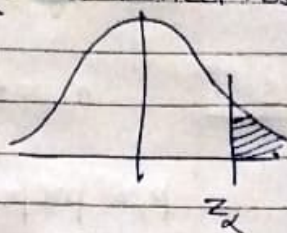
Null Hyp H_0 : let there is no significant difference b/w the consumption of tea before and after the increase in excise duty.

$$H_0: p_1 = p_2$$

Alternate Hypothesis:-

$$H_1: p_1 > p_2$$

(right tailed test)



$$\text{Test statistic } Z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$Z = \frac{0.8 - 0.667}{\sqrt{0.727 \times 0.273 \left(\frac{1}{1000} + \frac{1}{1200} \right)}}$$

$$Z = \frac{0.8 - 0.667}{0.019} = 7$$

$$\alpha = 5\%$$

$$Z_\alpha = -1.64$$

$$\text{Now } |Z_{\text{cal}}| > |Z_\alpha|$$

∴ null Hypothesis is rejected.
ie there is significant difference
in the consumption of tea before
and after the increase in excise duty.

Q: A machine produced 20 defective articles in a batch of 400. After overhauling it produced 10 defective articles in a batch of 300. Has the machine improved?

Solⁿ $n_1 = 400, \quad n_2 = 300$

$p_1 =$ proportion of defective item before overhauling
 $p_1 = \frac{20}{400} = 0.05$

$p_2 =$ proportion of defective item after overhauling
 $p_2 = \frac{10}{300} = 0.033$

$$\bar{p} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

$$\bar{p} = 0.043$$

$$Q = 0.957$$

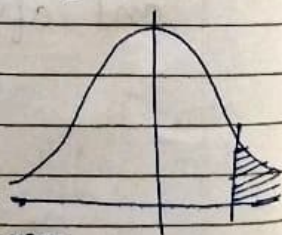
Null Hypothesis

(1) $H_0: p_1 = p_2$ i.e. proportion of defective items before and after overhauling are equal

(2) $H_1: p_1 > p_2$ (right-tailed test)

(3) level of significance $\alpha = 5\%$
 $= 0.05$

(4) Test statistic $Z = \frac{p_1 - p_2}{\sqrt{\bar{p}Q\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = 1.097$



$$Z_{\alpha} = 1.645$$

$$\text{as } |Z_{\text{cal}}| < |Z_{\alpha}|$$

ie Null Hypothesis is accepted at 5% level of significance ie proportion of defective item before and after are equal ie machine has not improved.

Q. A machine puts out 16 imperfect articles in samples of 500. After the machine is overhauled, it puts out 3 imperfect articles in a batch of 100. Has the machine improved?

Q'. 15.5% of a random sample of 1600 undergraduates were smokers, whereas 20% of a random sample of 900 postgraduates were smokers in a state. Can we conclude that less number of undergraduates are smokers than the postgraduates?

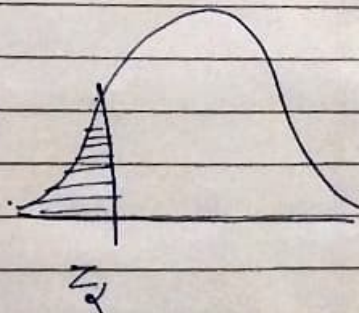
Solution:- $p_1 = 0.155$, $n_1 = 1600$
 $p_2 = 0.2$, $n_2 = 900$

Null Hypothesis $H_0: p_1 = p_2$

Alternate Hypothesis $p_1 < p_2$ (left tailed)

$$P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

$$P = 0.1712$$



Test Statistic

$$z = \frac{p_1 - p_2}{\sqrt{pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$z = \frac{0.155 - 0.2}{\sqrt{0.1712 \times 0.8288 \left(\frac{1}{1600} + \frac{1}{900}\right)}}$$

$$z = -2.87$$

$$\alpha = 5\% \text{ ie } z_{\alpha} = -1.645$$

$$\text{Now } |z_{\text{cal}}| \gg |z_{\alpha}|$$

ie Null Hypo is rejected. ie difference b/w p_1 and p_2 is significant.

ie H_1 accepted.

ie The Habit of Smoking is less among the undergraduate than among the postgraduates.